

5.1 Solving Multi-Step Equations

Lesson Introduction

 Benchmark	MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.		 Learning Targets	I can solve multi-step linear equations.
 Benchmark Coherence	Building On: MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context where all terms are rational numbers.			Working Toward: N/A
 Essential Question	What is the process for solving multi-step linear equations?			
 Lesson Narrative	In this lesson, students will extend their learning of solving basic one- and two-step equations from prior grade levels to solving multi-step equations. Students will collaborate to review the multiple approaches that can be taken to solve a multi-step equation and arrive at the same final answer. They will get ample opportunity to practice solving these types of equations both collaboratively and independently.			
 Component Summary	5.1.1 Warm-Up (~5 min.)	Aviation career exploration (SE p. 231)	5.1.4 Guided Practice (10-15 min.)	Solving multi-step equation practice problems (SE p. 234)
	5.1.2 Refresher (5 min.)	Solving two-step equations (SE p. 232)	5.1.5 Your Turn! (10-15 Min)	Independent practice solving multi-step equations (SE p. 234)
	5.1.3 Collaboration (10-15 min.)	Solving multi-step equations Collaboration activity (SE p. 233)	5.1.6 Wrap-Up (~5 min.)	Ticket Out the Door on solving multi-step equations (SE p. 235)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Warm-Up 5.1.1 contains a video to accompany the question prompts. (Optional) Calculators	N/A

 Teacher Tips	Common Misconceptions - Students may combine like terms across the equal sign instead of inverse operations. - Students may not distribute to every term when using the Distributive Property.	Things to Consider 5.1.2 Refresher serves as a reminder of solving two-step equations. Be cautious not to let this activity run too long during the class period, as it is below grade level and should serve as a refresher, not the primary core instruction.
	Literacy and Language Routines Think-Pair-Share 5.1.3 Collaboration leverages the Think-Pair-Share instructional routine in its design. Consider specifying for students the need for individual think time with the problem before beginning to work with their partner.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure Students learned how to solve one- and two-step equations in earlier grade levels. They will build upon previously learned concepts to relate prior learning to this new content.
	 Differentiate Instruction	
On-Ramp to Algebra 1 offers a variety of scaffolding and practice for students who need additional support with number sense, order of operations, and solving equations. Consider assigning an On-Ramp video and practice as “pre-work” for the lesson, even saving time by minimizing the need for 5.1.2 Refresher. - Number and Quantity: Multiplying an Integer by a Fraction - Algebra: Solving Equations (multiple videos), Multiplication Property of Equality, or Verifying Solutions of Algebraic Equations		
Lesson Notes:		

5.1.1 Warm-Up: Aviation Careers

Component Overview: Students explore careers in aviation by reading the passage below or watching the accompanying *Careers in Aviation* video.



5.1.1 Warm-Up: Aviation Careers

Aviation careers include pilots, air-traffic controllers, airport operators, airfield maintenance, aircraft rescue and firefighting, law enforcement and security officers, and project management.



1. Which of these careers in aviation had you not heard of before?

Answers will vary. I have never heard of an air-traffic controller.

2. Have you ever thought about a career in aviation?

Answers will vary. I have thought about being a pilot before, it seems like a great way to travel the world.



Consider increasing engagement in this activity by finding someone in the aviation industry to speak to the class in person or virtually.

5.1.2 Refresher: Solving Two-Step Equations

Component Overview: Students will engage in a refresher on solving two-step equations with three practice questions.



5.1.2 Refresher: Solving Two-Step Equations

1. Work with your partner to complete the statements using the words from the bank. Words may be used more than once.

coefficient	constant	variable
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- a. In the expression $7x - 2$, x is the variable, 7 is a coefficient, and -2 is a constant.
 - b. When solving an equation, the goal is to isolate the variable.
2. Solve each equation and verify your solution using substitution.
 - a. $8.25 - 1.5y = 1.8$
 $-1.5y = -6.45$
 $y = 4.3$

 $8.25 - 1.5(4.3) = 1.8$
 $8.25 - 6.45 = 1.8$
 $1.8 = 1.8$
 - b. $21 = 36\left(\frac{2}{3}a + \frac{5}{4}\right)$
 $21 = 24a + 45$
 $-24 = 24a$
 $-1 = a$

 $21 = 36\left(\frac{2}{3}(-1) + \frac{5}{4}\right)$
 $21 = 36\left(-\frac{2}{3} + \frac{5}{4}\right)$
 $21 = -24 + 45$
 $21 = 21$
 3. Compare your work with your partner's. Summarize any similarities and/or differences in your solution methods.
Answers will vary.



Be mindful of timing. Spending more than 5-8 minutes on the Refresher will take away instructional time from the primary learning targets. Instead, utilize this activity as a brief glimpse into determining which students may need additional support during Guided Practice.

5.1.3 Collaboration: Solving Multi-Step Equations

Component Overview: Students will collaborate on solving multi-step equations by exploring the work of different students. For this activity, students will need a partner.



5.1.3 Collaboration: Solving Multi-Step Equations

Javier and Arnav solved the same two equations, but they used different first steps.

1. In the table, **each student's first step** for solving each equation is started in blue. Work with your partner to finish solving the equations, continuing from each student's first step.

Javier	Arnav
$47 - 5.7m = 2.3m + 52$ $+ 5.7m \quad + 5.7m$ $47 = 8m + 52$ $-5 = 8m$ $-\frac{5}{8} = m$	$47 - 5.7m = 2.3m + 52$ $-47 \quad -47$ $-5.7m = 2.3m + 5$ $-8m = 5$ $m = -\frac{5}{8}$
$-\frac{1}{4}(28p + 60) = -6p + 1$ $(-\frac{1}{4})(28p) + (-\frac{1}{4})(60) = -6p + 1$ $-7p - 15 = -6p + 1$ $-15 = p + 1$ $-16 = p$	$-\frac{1}{4}(28p + 60) = -6p + 1$ $\frac{1}{4} \quad \frac{1}{4}$ $28p + 60 = 24p - 4$ $4p + 60 = -4$ $4p = -64$ $p = -16$

2. Discuss with your partner why both students had the same solutions for each equation, even though they used different strategies.
3. Discuss which strategies you preferred to start with when solving each of these equations. Summarize your discussion.

Answers will vary. They kept the equation balanced at each step, although they took different approaches.

Answers will vary. I preferred Javier's method in each equation because it created lesson room for errors for the way I approach the problems.



- What did Javier do differently than Arnav?
- What do you think that each student was thinking when they took the first step in each equation?



At the conclusion of the activity, engage students in a collaborative discussion as a whole group. Allow multiple students to share out to get varying perspectives and approaches.

5.1.4 Guided Practice: Solving Multi-Step Equations

Component Overview: Students will complete a Guided Practice activity with three multi-step equations.



5.1.4 Guided Practice: Solving Multi-Step Equations

- Solve each equation and verify the solution using substitution.

a. $8 + 5(16 + 9g) = 35g$
 $8 + 80 + 45g = 35g$
 $88 = -10g$
 $-8.8 = g$

$$8 + 5(16 + 9(-8.8)) = 35(-8.8)$$

$$8 + 5(16 - 79.2) = 35(-8.8)$$

$$8 + 5(-63.2) = -308$$

$$8 - 316 = -308$$

$$-308 = -308$$

b. $49 - 6d = \frac{3}{2}(28d - 10)$
 $49 - 6d = 42d - 15$
 $49 = 48d - 15$
 $64 = 48d$
 $\frac{4}{3} = d$
 $49 - 6\left(\frac{4}{3}\right) = \frac{3}{2}\left(28\left(\frac{4}{3}\right) - 10\right)$
 $49 - 8 = \frac{3}{2}\left(27\frac{1}{3}\right)$
 $41 = 41$

c. $\frac{62k-27}{11} = 8k + 7$
 $62k - 27 = 88k + 77$
 $26k = -104$
 $k = -4$

$$\frac{62(-4) - 27}{11} = 8(-4) + 7$$

$$-25 = -25$$



Consider your classroom when planning Guided Practice and Your Turn! Do your students need one more exemplar model? Is the classroom ready to have a student lead using inquiry? Should students work in pairs for question 1a and 1b, compare answers, then work independently on 1c before trying Your Turn! on their own?



For both 1b and 1c, consider highlighting differing strategies that students showcase. For example, some students could choose to multiply all values by the denominator of the fraction. Highlighting this strategy may contribute to more effective results in the Wrap-Up.

5.1.5 Your Turn!

Component Overview: Students have an opportunity to independently apply their learning to solve multi-step equations.



5.1.5 Your Turn!

1. Solve each equation.

a. $4.7y + 1.6 - 0.2y = 6.6$

$$4.7y - 0.2y = 5$$

$$4.5y = 5$$

$$y = \frac{10}{9}$$

b. $\frac{8}{3} + 29z + 1 = 7z - \frac{1}{3}$

$$29z + 1 = 7z - 3$$

$$22z = -4$$

$$z = -\frac{2}{11}$$



Consider pulling a small group of students who struggled with the Guided Practice activity. Differentiate the instruction to target conceptual understanding rather than errors in calculation.



For students who finish early, challenge them to create a multi-step problem and trade with a partner to solve each other's equation.

5.1.6 Wrap-Up: Cool Down

Component Overview: Students will complete a Ticket Out the Door with two multi-step equations.



5.1.6 Wrap-Up: Cool Down

- Solve each equation and check the solution using substitution.

a. $23.5 - 32p = 16.5p + 4.1$

$p = 0.4$

$23.5 - 32(0.4) = 16.5(0.4) + 4.1$

$23.5 - 12.8 = 6.6 + 4.1$

$10.7 = 10.7$

b. $\frac{3}{4}(12n - 10) = 19n - 3.5$

$n = -0.4$

$\frac{3}{4}(12(-0.4) - 10) = 19(-0.4) - 3.5$

$\frac{3}{4}(-14.8) = -11.1$

$-11.1 = -11.1$



Monitor for potential common errors that could inform small groups in Lesson 2:

- Students may add 23.5 to both sides, incorrectly associating the subtraction sign with 23.5.
- Students may add 32 to both sides, treating the -32 as an integer rather than a coefficient.
- Students may mix their tenths and hundredths places when combining like terms.



Some students may distribute $\frac{3}{4}$ first, while other students may multiply all values by 4 to reduce the denominator. Look for students who do both and consider spiraling their work with the fraction in the next lesson's Collaboration activity.